

# Esa Fallah Royani

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## Summary

Electrical Engineering graduate focused on electronics, control systems, and sustainable technology, with particular experience in DC-DC converter design and development. Experienced through research, internships, and engineering projects involving electronic hardware, electrical panels, PLC programming, measurement, and testing. Interested in developing practical engineering solutions, particularly in power electronics and energy-related applications.

## Education

- **Institut Teknologi Nasional Bandung**  
Bachelor of Electrical Engineering | 2019–2022  
GPA: 3.33 / 4.00
  - Achievements
    - 2nd Place Project Innovation Competition, Sunan Gunung Djati Robotic Competition, 2023
    - 3rd Place, University Level Appropriate Technology Competition, 2024
    - Silver Medal Mercu Buana Innovation Week, 2024
    - Bronze Medal Mercu Buana Innovation Week, 2025
    - Semifinalist Samsung Solve for Tomorrow, 2024
    - Presenter 6th International Conference on Green Technology and Design, 2024
- **SMK Negeri 4 Bandung**  
Audio Video Engineering | 2019–2022
  - Achievements
    - 1st Place — Provincial Electronics Student Competency Competition (LKS), 2021
    - Medallion for Excellence — National Electronics Competition, 2021

## Professional Experience

- **Research Intern – Pusat Teknologi Transportasi, BRIN**  
1 November 2025 – 8 August 2026
  - Worked on the development of a DC-DC charging system for the receiver side of a Wireless Power Transfer (WPT) system for electric vehicle battery charging.
  - Worked on electronic control systems, measurements, Simulation and testing related to the research project.
  - Applied control system and power electronics concepts in the development and evaluation of the converter system.

- **PLC Programmer Intern – PT. Renerconsys**  
23 Oktober 2024 – 23 November 2024
  - Developed a PLC simulation program using Mitsubishi GX Works2 for a trashrack cleaner automation system.
  - Applied basic PLC programming and industrial automation concepts during the development process.
- **Production Team — Ventilator Indonesia (Vent-I)**  
*Maret 2020 - April 2020*
  - Identified and checked passive and active electronic components.
  - Used multimeters and oscilloscopes for basic voltage, current, resistance, and frequency measurements.
  - Performed manual PCB component assembly and soldering.
  - Conducted PCB assembly quality checks and functional testing.
  - Assisted with hardware assembly and testing of Vent-I units.
- **Engineering Electrical Panel Intern – PT. AEMCO Persada Nusantara**  
*April 2021 - Agustus 2021*
  - Performed quality checks on electrical components before panel assembly.
  - Read technical drawings for electrical panel installation and wiring.
  - Installed and wired electrical components according to the provided installation drawings.
  - Performed panel inspection, measurement, and functional testing.
- **Uji Kompetensi Keahlian Teknik Audio Video SMK Negeri 4 Bandung**  
*April 2022*
  - Identified passive and active electronic components.
  - Performed basic measurements using multimeters and oscilloscopes.
  - Performed PCB assembly and manual soldering.
  - Conducted PCB assembly quality checks and functional testing.

## TECHNICAL SKILLS

- **Electronics & Hardware**
  - Electronic Schematic Design — Autodesk Eagle
  - Electronic Circuit Simulation — Proteus, Ltspice
  - Electronic Hardware Prototyping
  - PCB Assembly & Soldering
- **Measurement & Testing**
  - Mitsubishi GX Works2
  - Mitsubishi GT Designer
  - PLC Programming & Simulation

- Basic Control System Concepts
- C++ & Arduino

## **SOFT SKILLS**

- Problem Solving
- Collaboration
- Communication
- Willingness to Learn

## **LANGUAGES**

- Indonesian: Native
- English: Conversational

## **Certifications**

- **Lembaga Sertifikasi Profesi Elektronika Indonesia (LSP-EI)**  
Competency Certification - BNSP  
*Juni 2022*